

DAVIDE ITALO SERRAMAZZA

1 ROEBUCK HOUSE, Dublin, Ireland, +39 331 569 7012

davide.serramazza@gmail.com — <https://davide-serramazza.github.io/>

github.com/davide-serramazza — [linkedin.com/in/davide-italo-serramazza-88b9501a2/](https://www.linkedin.com/in/davide-italo-serramazza-88b9501a2/)

Professional Summary

ML scientist with a PhD in Explainable AI (thesis submitted, viva pending) and 5+ years of experience building production ML systems. Author of open-source projects (tsCaptum on PyPI, drXAI on GitHub), 4 first-author papers at *ECML-PKDD* and 1 journal article. Seeking Applied/Research Scientist and ML Engineer roles where I can solve ML problems from research to production.

Skills

- **Programming Languages:** Python, Node.js, MATLAB, Java, JavaScript, C++, C, Bash scripts
- **ML/DL frameworks:** PyTorch, TensorFlow, sktime, aeon, scikit-learn, Captum
- **XAI:** SHAP, KernelSHAP, LIME, Feature Ablation, Integrated Gradients, GradientSHAP, CAM
- **Data Handling:** pandas, numpy, SQL, MongoDB, familiar with Spark
- **Software engineering:** OOP, design patterns, API design
- **Visualization:** matplotlib, seaborn
- **Other:** Linux, MongoDB, FFmpeg, REST APIs, CUDA acceleration

Experience

University College Dublin

September 2022 – Present (thesis submitted; viva pending)

PhD in AI for Time Series

Dublin, Ireland

- Adapted SOTA XAI methods (SHAP, LIME, Feature Ablation) to CNN, transformer, and time-series classifiers (ROCKET, etc), **improving saliency map accuracy while cutting computation time**.
- Built tsCaptum, a library that reduces XAI computation time for time-series models **by 10–100× without degrading attribution quality**. Includes customizable batching, background selection, and visualization primitives.
- Developed drXAI, a feature-selection framework that **discards 80–90% of input data while preserving model accuracy**, cutting inference cost and data pipeline load.
- Led **4 first-author research projects** published in *ECML-PKDD* + 1 article in *Software impact*. [Google scholar link](#)

University of Pisa (with Mediateca Toscana)

May 2020 – May 2022

Research Fellowship / ML Engineer (Assegno di Ricerca)

Tuscany, Italy

- Built a **production ready** end-to-end ML pipeline: data ingestion → model training/inference → clip extractions → deployment (API + database). Code: github.com/Unipisa/PH-Remix-extractor-processor-spooler — github.com/Unipisa/PH-Remix-restAPI
- Processed 100+ hours of documentary footage with deep learning models, enabling a web platform for clip labeling, querying, and remix creation.
- Worked directly with domain experts and client stakeholders to translate requirements into a functional system.
- *Technical stack:* Python, PyTorch, TensorFlow, GitHub, Node.js, MongoDB, FFmpeg, REST APIs

Selected projects

- **tsCaptum** Python library for XAI on time series. PyPI pypi.org/project/tsCaptum — GitHub: github.com/davide-serramazza/tscaptum
- **drXAI** Feature-selection framework for time-series models. GitHub: github.com/davide-serramazza/drXAI

Education

Master in AI (School of Computer Science), Grade 110/110

September 2017 – March 2020

University of Pisa

Pisa, Italy

Bachelor in Computer Science

September 2013 – September 2017

University of Pisa

Pisa, Italy

Languages

- **Italian** (native)
- **English** (Professional working proficiency)